JINGXUAN DUAN

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EDUCATION

Carnegie Mellon University Aug 2025 – Dec 2026

M.S. in Civil and Environmental Engineering GPA: 3.87/4.0

Track: Sustainable Energy and Transportation Systems (SETS).

Core Courses: Machine Learning (10-601); Deep Reinforcement Learning (10-703); Computer Vision (16-720); Visual Learning and Recognition (16-724); F1TENTH (16-664); Probabilistic Estimation Methods for Engineering Systems (12-704); Sensors, Data, and Models in Engineering Systems (12-778).

Southeast University Aug 2021 – Jun 2025

B.Eng. in Highway and Bridge Engineering GPA: 85.6/100

Core Courses: Advanced Mathematics; Linear Algebra; Structural Mechanics; Mechanics of Materials; Theoretical Mechanics.

University of California, Los Angeles Aug – Sep 2023

Summer Session GPA: 4.0/4.0

RESEARCH INTERESTS

Multimodal fusion and computer vision for perception and behavior understanding; human-computer interaction and human skill capture; task-aware representations and world models for data-efficient robot learning; perception and control for autonomous systems in unstructured environments.

RESEARCH EXPERIENCE

TGIF: Multimodal Temporal Segmentation Summer – Sep 2026

Summer Research, Department of Civil and Environmental Engineering, Carnegie Mellon University

Advisor: Prof. Pingbo Tang

- Improved TGIF video segmentation with dual-ROI VideoMAE/MS-TCN, raising validation accuracy from about 80% to 91.9%. Multimodal fusion reached 92.5% mean accuracy (93.7% best seed) and improved Edit from 89.7% to 91.4%.
- Extended Dynamic Window Attention (DWA) for WEAR with AC-RMS envelopes and ordered slope histograms to resize sensor windows before one-way video-to-sensor attention and two-round shared MS-TCN refinement.
- On WEAR (18-subject LOSO, seed 47; fixed final checkpoints), achieved **75.14% subject Macro-F1** and **71.44% mAP@0.5** (+1.10/+1.54 pp over previous DWA). Cut single-thread CPU forward time by **40.2% (1.67x)** on three sequences, after feature extraction.

Autonomous Delivery Robot System Jan – May 2026

Independent Study, Department of Civil and Environmental Engineering, Carnegie Mellon University

Advisor: Prof. Pingbo Tang

- Formulated a multi-trip, multi-commodity vehicle-routing model with trip-indexed routes, explicit unloading variables, and flow conservation to enforce capacity and depot-loading consistency. Introduced explicit lower and upper trip-count bounds to constrain the CBC solver's search space.
- Built a three-floor delivery system linking semantic topology, Theta* planning, CBC/OR-Tools scheduling, and Unity/C# execution across ramps and an elevator; completed 100/100 randomized simulated deliveries with a mean round-trip time of 223 s. Evaluated SAC-based local recovery in Python with task-specific rewards for obstacle avoidance and route rejoining.

RESEARCH EXPERIENCE (CONTINUED)

DELTA: Task-Aware Anchor-Delta Representations Sep 2026 – Present

Ongoing Research in Robot Learning

- Investigating task-aware visual anchors and action-conditioned deltas that preserve decision-relevant information for manipulation policies without reconstructing dense video.
- Designing a LeRobot-compatible pipeline with delta-triggered anchor refreshes; planned experiments assess storage, loading throughput, and policy success against dense-video and sampling baselines.

DMI Eye-Tracking: Manufacturing Skill Capture Sep 2026 – Present

Early-stage Research, MFI Project, CMU

- Investigating how eye-tracking video, operator narration, and machine context can capture tacit knowledge in roll-forming changeovers; focusing on quality-inspection decisions and Observe–Orient–Decide–Act (OODA) episodes for operator guidance.

INTERNSHIP EXPERIENCE

ByteDance Jan 2022; Dec 2022 – Jan 2023; Oct 2024 – Jan 2025

Data Analyst Intern, Real Estate & CIS (Corporate Information Systems), Beijing / Shanghai

- Analyzed global workspace utilization and return-to-office trends; maintained workstation-, zone-, and region-level metrics to support space planning and leasing decisions.
- Built an interactive dashboard of nationwide workstation changes using Lark and Excel; managed data access and supported cross-regional reporting.
- Conducted sentiment and topic analysis for workspace operations and competitive product evaluations for height-adjustable desks.

BYD, Automotive Engineering Research Institute Jan 2024

Intern, Xi'an

- Built a vehicle simulation framework and powertrain modules in AVL CRUISE for the European-export Song PLUS; optimized fuel economy and vehicle performance through simulation testing under NEDC, WLTP, and CLTC driving cycles.
- Conducted midsize SUV competitor analysis and contributed to a product positioning manual.

PUBLICATION, HONORS & AWARDS

Publication: C. Chen, **J. Duan**, S. Zou, S. Chen, and X. Huang. "Study on skid resistance of thin-layer porous asphalt pavement." *Modern Transportation and Metallurgical Materials*, 3(1): 2–8, 2023 (in Chinese).

Southeast University: Outstanding Graduate and Outstanding Graduation Thesis (2025); Merit Student (2025); Zhishan Student Scholarship (2023, 2025).

Competitions: National 1st Outstanding Winner, Mao Yisheng Public Bridge Design Competition (2025); Gold Award, SEU "Celebration Cup" Innovation and Entrepreneurship Plan Competition (2024); Honorable Mention, Mathematical Contest in Modeling (2024).

SKILLS & INTERESTS

Technical: Python, AI Agent Tools, SQL, ROS, Unity, Revit, Microsoft, C#, MATLAB.

Interests: Choral singing (tenor, SEU Chamber Choir); clarinet; football (SEU and School of Transportation teams); basketball; badminton.